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Optimizing support vector regression using grey wolf optimizer for enhancing energy efficiency and building prototype architecture

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Abstract

In this era of increasing energy demand, optimizing energy consumption in building systems is critical for enhancing sustainability and operational efficiency. Residential buildings consume significant energy, with heating load (HL) and cooling load (CL) being major contributors. While previous research explored modeling techniques, there's a need for more precise and optimized approaches. It has been observed that accurately predicting heating and cooling load can help in designing better building prototypes with improved energy efficiency. Researchers have explored multiple computational intelligence approaches that have improved the prediction of energy consumption of buildings. However, the applied approaches have some limitations. To improve the prediction of HL and CL of buildings and to overcome the limitations of the existing approaches, this study employs a support vector regression (SVR) fine-tuned with Grey Wolf Optimization (GWO) for enhancing energy efficiency and building prototype architecture by predicting HL and CL. To establish the supremacy of the proposed approach experiments are conducted on a benchmark dataset and performance is compared with four prominent machine learning (ML) algorithms namely linear regression, decision tree, random forest, and deep neural networks to predict HL and CL. The results demonstrate that SVR-GWO outperforms the others, achieving an r-score of 99.77% for HL and 97.90% for CL. Additionally, various metrics like Mean Absolute Error (MAE), Mean Square Error (MSE), and Root Mean Square Error (RMSE) are computed to evaluate model performance. The HL model achieved MAE, MSE, and RMSE scores of 0.3422, 0.2354, and 0.4852, respectively. Similarly, the CL model achieved scores of 0.8387, 1.9426, and 1.3937, respectively. The results show better performance of the proposed model in comparison to the ML models.

Keywords Energy efficient building · SVM · Grey wolf optimization · Heating load · Cooling load · Forecasting

1 Introduction

The need for energy-efficient buildings has shown significant and rapid growth in recent years. It now become a fast-moving movement around the globe. There are multiple justifications to endorse this movement. The need to address global warming, energy problems, acid rain, water pollution, and other environmental issues has encouraged a study on the energy efficiency of buildings [1–3]. Currently, Leadership in Energy and Environmental Design (LEED) is considered the most appropriate rating system developed by the Green Building Council (USGBC) of the United States [4]. It guides the construction of environmentally friendly and sustainable structures. Recently, there has been a significant increase in the publication of books, reports, and studies that discuss ideas and designs aimed at improving building energy efficiency [5, 6].

The World Business Council for Sustainable Development reports that buildings account for a substantial portion (around 40%) of the total energy used by many nations [2]. The International Energy Agency predicts that without alternative green energy sources, three-fourths of all energy will continue to be associated with carbon dioxide by 2030.

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Nowadays, people are focusing on creating buildings that use less energy by incorporating sustainable design principles [7]. Moreover, according to European Directive 2002/91/EC, buildings in European countries are required to follow the rules to meet adequate minimum energy efficiency requirements [8, 9]. The report shows that in the past few decades, building energy consumption around the world has substantially increased. HVAC (Heating, Ventilating, and Air Conditioning) is a system that provides heating and cooling to large commercial buildings. It acts as a catalyst in the regulation of the interior climate [10, 11]. As a result, the design of more energy-efficient buildings with higher energy-saving capabilities is a strategy for reducing the ever-increasing demand for additional energy supplies [12]. To determine the necessary heating and cooling equipment for maintaining comfortable indoor temperatures, it is essential to calculate the heating load (HL) and cooling load (CL) [13]. To evaluate the required heating and cooling capacity, building designers and architects need to find all the information about a building and its characteristics. Information about space (activity level and occupancy), the effect of climate, and the use of buildings [14]. Numerous publications, magazines, and books on architectural design and energy efficiency have highlighted the characteristics that contribute to a building's energy efficiency [15, 16]. Energy-efficient homes incorporate features such as high-performance windows, enhanced insulation, airtight construction, sealed ducts, efficient heating and cooling systems, and energy-saving appliances. These homes often employ architectural designs that minimize energy consumption. A thermodiode system can be implemented to address the issue of excessive solar heat gain in summer. The authors in [6] developed bi-directional thermodiodes that can minimize solar heating in summer and during the winter, it can be used to generate heat. People have now become aware of the negative consequences of excessive energy consumption in buildings [17]. As a result, interest among researchers in the design of sustainable buildings has increased. Buildings developed using sustainable architecture are referred to as green buildings and this architecture aims to reduce the impacts of construction on the environment. However, unless it has been rated by the Building Research Establishment Environmental Assessment Method (BREEAM) or LEED, it will not be considered a green building [18]. To develop green buildings with sustainable architectures, it is essential to predict the HL and CL accurately. Mismanagement in predicting these loads can lead to inefficient energy use, excessive costs, and increased carbon footprints. Several studies have demonstrated the benefits of accurate forecasting for energy efficiency, with notable works [19]. These studies highlight the need for robust prediction models to handle seasonal variations, building types, and climatic conditions[20, 21]. There are multiple prediction techniques ranging from simple linear regression to advanced machine learning used by researchers to predict HL and CL of buildings [22]. A summary of different prediction techniques that have been used for predicting HL and CL is provided in the related work section. These techniques have their limitations and can be improved in many directions. More accurate prediction methods are essential to enhance energy efficiency and sustainability.

this direction, this paper proposes an efficient model to predict and enhance the energy efficiency of buildings and their prototype architecture. The paper applies support vector regression fine-tuned with grey wolf optimizer (SVR-GWO) to enhance the energy efficiency and improve the building prototype by predicting HL and CL. The flow diagram showing the organization of different sections of the paper is shown in Fig. 1. Table 1 presents the nomenclature of the paper. The key contributions to this paper are as follows:

- Proposed SVR_GWO model for predicting HL and CL of buildings and enhancing building energy efficiency and prototype architecture.
- The proposed SVR-GWO model achieved the highest coefficient of determination (R^2) for both heating and cooling load prediction compared to other benchmark models (DNN, RF, DT, and SVR-Grid).

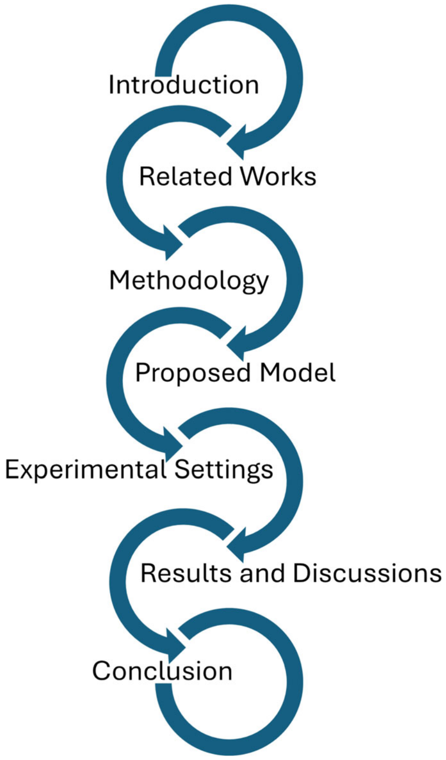

Fig. 1 Flow of the paper

Table 1 Nomenclature of the paper

Abbreviation	Description
SVR	Support vector regression
GWO	Grey wolf optimization
MSE	Mean square error
DL	Deep learning
ML	Machine learning
MAE	Mean absolute error
DT	Decision tree
RF	Random forest
DNN	Deep neural network
LEED	Leadership in energy and environmental design
HVAC	Heating, ventilating, and air conditioning
HL	Heating load
CL	Cooling load
BREEAM	Building research establishment environmental assessment method
RBF	Radial basis function
SVM	Support vector machine
RMSE	Root mean square error
LR	Linear regression
ANN	Artificial neural network
UCI-ML	University of California Irvine, Machine Learning
SCE	Shuffled complex evolution
MLP	Multi player perceptron
GBM	Gradient boosted machines
GPR	Gaussian process regression
UDI	Useful daylight illuminance
DA	Daylight autonomy
SDA	Spatial daylight autonomy
ASE	Annual solar exposure
CFD	Computational fluid dynamics
TRNSYS	Transient system simulation tool
IBE-e	Integrated building environment-environment
CCHP	Combined cooling, heating, and power
DeST	Designer’s simulation toolkit
IES-VE	Integrated environmental solutions virtual environment
CONTAM	Contaminant transport analysis
AHU	Air handling unit
VAV	Variable air volume
SD	Standard deviation
NC-	Not calculated
LM-MLP	Levenberg–Marquardt (LM-MLP)
MFO	Moth-flame-optimization
OIO	Optics inspired optimization
SCE	Shuffled complex evolution
PSO	Particle swarm optimization

- The paper highlights that using GWO for parameter optimization in SVR led to significant improvement in performance compared to other optimizers like Grid Search.
- The study observed that Deep Neural Network (DNN) underperformed compared to other models, possibly due to the limited size of the training data.

The remaining sections of this paper are organized as follows. Section 2 presents the related works, Sect. 3 discusses the methodologies. The proposed model is elaborated in Sect. 4. After that, Sect. 5 discusses the experimental settings. Section 6 is devoted to results and discussions. Finally, Sect. 7 is the conclusion.

2 Related works

Accurate prediction of heating and cooling loads plays a crucial role in designing energy-efficient buildings, with numerous studies focusing on this domain. Different machine learning models and optimization strategies have been used to enhance prediction accuracy. One of the most commonly explored datasets is the UCI Energy Efficiency Dataset, known for its use in several landmark studies on building energy efficiency prediction.

Multiple Regression Analysis is one of the early approaches for forecasting HL. It simplifies the prediction process but struggles to address the non-linear patterns present in the data [23]. To overcome this hurdle, Artificial Neural Networks have been investigated for their capacity to identify intricate, nonlinear patterns. In this regard, [24] assessed the capability of ANN to predict CL and HL, particularly about thermal storage. While ANN shows improved accuracy, its computational complexity became a concern, leading researchers to explore alternative methods.

One such method is the Support Vector Machine (SVM), which introduces a novel aspect to building energy prediction. However, its early implementations, as shown in [25], demonstrated lower accuracy compared to ANN, primarily due to the absence of optimization in the model's parameters. Studies like [26] revealed that SVM can outperform back-propagation-based ANN in predicting cooling loads, especially in regions like Guangzhou, China. However, this study did not delve into the crucial aspect of hyperparameter optimization, which is key to enhancing model performance.

This gap in optimization is where recent studies have made significant strides. In a recent study, Lu et al. [27] presented an automated machine-learning approach that can efficiently construct machine-learning pipelines tailored to HL and CL forecasting. This framework significantly outperformed traditional models, achieving R^2 values of 0.9965 for HL and 0.9899 for CL predictions. Unlike earlier models, AutoML reduces the need for expert manual intervention, particularly during the early stages of building design, marking a shift toward more autonomous systems. Similarly, Bashir et al. [28] reviewed all the recent models that have focused on accuracy and model performance to forecast the heating and cooling load.

Another major advancement comes from Zheng et al. [29], who introduced a hybrid ML approach using Shuffled Complex Evolution (SCE) to optimize a multi-layer perceptron neural network (MLP). Their work demonstrated that SCE-MLP improved prediction accuracy by 22.84% over traditional MLPs while also reducing computational time. They highlight the power of metaheuristic optimization techniques, showcasing their ability to enhance ML models' predictive performance in energy-efficient buildings.

Roy et al. [30] took a broader approach by comparing various ML models, including Deep Neural Networks (DNN), Gradient Boosted Machines (GBM), and Gaussian Process Regression (GPR). Their results showed that DNN and GPR outperformed traditional models, achieving VAF scores exceeding 99% for both heating and cooling load. These findings suggest that advanced models like DNN should be considered over simpler methods like unoptimized SVM, as they can better capture the intricate relationships within energy datasets.

Similarly, Gao et al. [31] conducted a comprehensive analysis using sixteen ML models to predict HL and CL. Their work emphasized the importance of ensemble methods like Random Forest and Alternating Model Trees, which were shown to outperform other models in both accuracy and robustness. This reinforces the growing reliance on ensemble methods and feature selection techniques to handle the complexity of energy efficiency data.

With rising pollution, environmental concerns, and a shortage of energy supplies, the need for energy-efficient buildings has become ever more important. The concerns around the issue have led to ample studies on enhancing the energy efficiency of building architecture, particularly in the heating and cooling systems, which use most of a building's energy. Fisk and William [32] say that the idea of energy-efficient buildings has been based on knowing how physical design affects energy use. New developments, like those discussed by Kehinde et al. [33], have added the use of advanced technologies and materials to make buildings even more energy efficient by using smart, adaptable materials. Parallel to these developments, the integration of ML into energy efficiency optimization represents a cutting-edge approach. Pham et al. [34] provided an early insight into the potential of ML in predicting energy consumption patterns in buildings, demonstrating basic ML models' capabilities in deciphering complex energy dynamics. The application of SVM in energy prediction, a specific area within ML, has garnered attention. Dong and Bing [35] applied SVM in energy consumption predictions for commercial buildings with promising outcomes. However, their research did not delve into the crucial aspect of hyperparameter optimization of SVM, which is essential for enhancing the accuracy of predictions.

Techniques like Grid Search Cross-Validation and Bayesian Search Cross-Validation are being used more and more to focus on SVM's hyperparameter optimization. In 2020, Oliveira et al. [36] used Grid Search Cross-Validation to improve the performance of SVMs for predicting financial markets. This showed that it is an effective way to improve model performance. In the same way, Laurensia et al. [37] showed that Bayesian Search Cross-Validation was better at optimizing SVM parameters for medical diagnostic uses. However, there isn't a lot of research that compares SVM to other ML and DL algorithms like DT, RF, and DNN, especially when it comes to how energy-efficient buildings are. In 2015, Kaytez [38] proposed SVM and linear regression in predicting energy use, showing that SVM was generally better. However, there is still a lack of study that compares SVM to other methods in more detail. Additionally, Bui et al. [39] developed a hybrid model combining M5 Rules with Genetic Algorithms (GA) to predict the heating load of buildings. The hybrid model demonstrated significant accuracy improvements over standalone models. This study showcases the effectiveness of rule-based systems in conjunction with evolutionary algorithms for better performance in energy-efficient building predictions.

The potential application of predictive models in the pre-construction phase of building design is a burgeoning area of interest. Mohamed and Nader [40] underscored the benefits of employing predictive models early in the design process, enabling architects to make informed decisions geared toward enhancing energy efficiency. This approach aligns with the current research, which aims to bridge the gap in the literature by applying an optimized SVM method to estimate energy efficiency in building prototypes. The study further extends to a comparative analysis with other prominent ML and deep learning algorithms, offering a new perspective on accurate and reliable energy consumption predictions, particularly for heating and cooling systems.

Table 2 summarizes the applications of performance-driven design in various building types, focusing on the different simulations and analyses used to optimize building performance. The provided table showcases how performance-driven design is applied across diverse building types. Various simulation tools are employed, with Rhinoceros/Grasshopper/Honeybee for offices and Design Builder/EnergyPlus for residences being common examples. Energy-related, thermal, and daylighting analyses are frequently conducted, while air quality analysis appears less widespread. Notably, most studies aim to optimize energy consumption, CO_2 emissions, and thermal comfort. While information is missing for some analyses in certain building types, the table demonstrates the potential of performance-driven design to optimize various aspects of building performance across different typologies.

Building simulations are increasingly utilized to optimize building operations across diverse typologies. Table 3 showcases various applications, from optimizing HVAC setpoints in educational buildings to predicting filter pressure drops in data centers. Optimization strategies range from rule-based control to genetic algorithms, targeting key systems like chillers, AHU, and ventilation networks. Ultimately, these approaches aim to achieve improved thermal comfort, reduced energy consumption, and optimized system performance, contributing to sustainable and efficient building operations.

3 Methodologies

3.1 Support vector regression

The SVR model is based on a linear regression analysis which can be mathematically represented as in Eq. (1).

$$f(x) = w^T x + b \tag{1}$$

where $w, x \in R^d$, and d represent the dimensionality of the input vector. Being a multinomial problem, many regression solutions are possible, however, optimal regression can be selected by minimizing the following function as described in Eq. (2)

$$\frac{1}{2} w^T w + \frac{1}{n} \sum_{i=1}^{n} c(f(x_i), y_i) \tag{2}$$

where $c(f(x_i), y_i)$ is the cost or loss function which decides the quantity of penalty needed to tune the weights (w) according to the difference received between $f(x_i)$ and y_i. It is depicted below in Eq. (3) if ε insensitive.

$$c(f(x_i), y_i) = \begin{cases} 0, & |y - f(x)| \le \varepsilon \\ |y - f(x)| - \varepsilon, & otherwise \end{cases} \tag{3}$$

From the above equation, it is clear to understand that SVR not only considers the points on the line but also generates a tube around the regression line of radius ε and takes the loss as null for all the points in the tube. Furthermore, for the point outside the tube difference between the radius and gap from prediction and actual value

Table 2 Applications of Performance-Driven Design

Building type	Simulation tool	Energy-related analysis	Thermal analysis	Daylighting analysis	Air quality analysis	Optimization objectives	Source
Office	Honeybee, Grasshopper, Rhinoceros	Annual solar exposure, Useful daylight illuminance, Spatial daylight autonomy	✔	✔	✔	Energy consumption, CO2 emissions, thermal comfort	[41]
Residential	DesignBuilder; EnergyPlus	CO2 emissions, Energy consumption	✔	✔	✔	✗	[42]
Hospital	CFD	Average air age	✔	✗	✔	✗	[43]
Office	TRNSYS; IBE-e	Energy consumption	✔	✔	✗	✗	[44]
Education	Radiance, Sketchup	Useful daylight illuminance, Daylight autonomy	✔	✔	✗	✗	[45]
Education	DesignBuilder; eQUEST; EnergyPlus	Indoor air temperature, CO2 concentration	✔	✔	✔	✗	[46]
Education	Radiance; EnergyPlus	Heating loads, and cooling loads, Daylighting scores,	✔	✔	✔	✗	[47]
Office & Education	EnergyPlus	Exergy destruction, Energy use, non-comfortable hours,	✔	✗	✗	✗	[48]
Office	Rhinoceros; DIVA	Useful daylight illuminance	✔	✔	✗	✗	[49]

becomes the loss. In this case, Eq. (1) can be minimized subject to the following constraints given in Eq. (4).

$$\begin{aligned} y_i - wx_i - b &\le \varepsilon + \xi_i \\ wx_i + b - y_i &\le \varepsilon + \xi_i^* \\ \xi_i \xi_i^* &\ge 0 \end{aligned} \tag{4}$$

where c is the positive constant, ε deviation, and ξ_i, ξ_i^* are slack variables.

3.2 Hyperparameters tuning

The regression model is adjusted using three primary settings (C, Gamma, and Kernel). These settings and their influence on the model are explained below.

Kernel: Kernel is a function of a hyperparameter which is used to create boundaries or curves to estimate the value by fitting values as shown in Fig. 2. There are many kernels developed and used by researchers like linear, nonlinear, or polynomial, Gaussian, etc. However, RBF is a well-known and highly effective kernel used for almost all kinds of problems [60]. Although RBF is an efficient kernel, it is hard to say which one reflects the best fit for a specific problem. Figure 3 is a clear demonstration of different kernels and shows how they differ. Therefore, we used linear, polynomial, and RBF during the initial stage and applied optimization algorithms. Linear and polynomial kernels are presented in Eqs. (5) and (6) respectively.

$$k(x_i, x) = b_0 + \sum_i a_i * (x, x_i) \tag{5}$$

$$k(x_i, x) = 1 + \sum_i (x * x_i)^d \tag{6}$$

After optimization, we found that RBF is the most suitable kernel for the proposed problem, and it can be mathematically demonstrated below as in Eq. (7) and the constraint given in Eq. (8).

$$k\left(x_i, x_j\right) = \exp\left(-\gamma x_i - x_j^2\right) \tag{7}$$

where

$$\gamma > 0 \quad \text{and} \quad \gamma = \frac{1}{2\sigma^2} \tag{8}$$

C: C is a hyperparameter used to define the wideness of the SVM kernel function. In other words, C is a coefficient that is employed to the maximum gap between data points and the boundary. Large C values leave less space between the boundary and data points however small C values create a wider space. $C \in \mathbb{R}^+$. Initially, for the proposed model, we have given many possible values for C and after applying optimization techniques we found the optimal one.

Gamma: To add nonlinearity in the SVM kernel function (Hyperplane) a coefficient known as Gamma is used. The high value of Gamma can add more non-linearity and fit a tight bond around the training data points, however

Table 3 Applications of Building Simulation for Operation

Source	Building type	Tools	Application (case study)	Optimization strategy	Key systems/components	Outcomes
[50]	Non-residential	Modelica	Compressor speed PID-controller parameters (KP & TI)	Model-based predictive control	HVAC system	Reduced energy consumption
[29]	Educational	Design Builder	Heating and cooling setpoint temperatures	Rule-based scheduling based on predicted occupancy	HVAC system	Improved thermal comfort and reduced energy use
[51]	Hotel	DeST	Solar VAV output schemes under climate change	Genetic algorithm	CCHP system	Optimized energy production and reduced emissions
[52]	Mosque	EnergyPlus	HVAC setpoint schedule based on thermal comfort and occupancy prediction	Model-predictive control with machine learning	HVAC system	Improved occupant comfort and reduced energy consumption
[53]	Educational	R	Chilled water flow rates, condensing water flow rates, Chilled water temperature and the speed of the cooling tower fan	Genetic algorithm	Chilled water system and cooling tower	Enhanced system efficiency and reduced energy costs
[54]	Metro station	TRNSYS	Chiller performance optimization through adjusting setpoints for chilled water outlet temperatures	Rule-based control with predictive optimization	Chiller system	Reduced chiller energy consumption
[55]	Commercial	IES-VE	Supply air temperature and VAV airflow	Multi-objective optimization	AHU and VAV system	Improved thermal comfort and energy savings
[56]	–	EnergyPlus, CONTAM, Matlab	Damper angles and fan pressure	Evolutionary algorithm	Ventilation system	Optimized airflow distribution and reduced energy use
[57]	–	Matlab	Predict pressure drops in AHU filters due to clogging	Machine learning	AHU filters	Proactive maintenance and reduced downtime
[58]	Datacentre	TRNSYS	Select optimal cooling mode (mechanical, partial, free)	Rule-based control with weather prediction	Cooling system	Minimized energy consumption while maintaining cooling requirements
[59]	Office	EnergyPlus	Window and ventilation fan control in mixed-mode buildings	Model-predictive control	Windows and ventilation system	Reduced energy use while maintaining indoor air quality

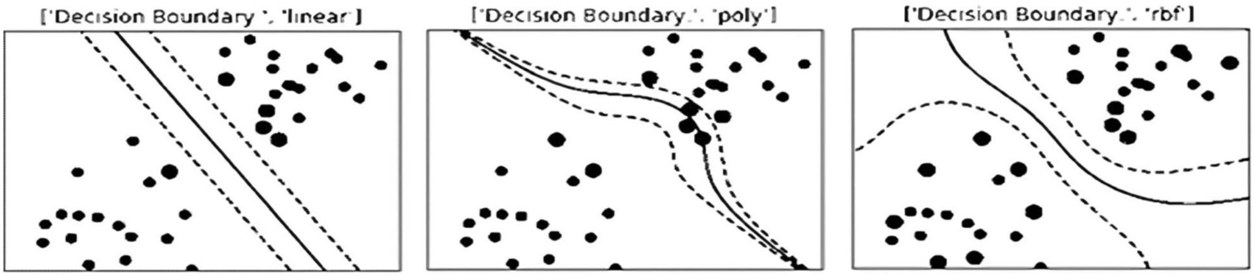
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Fig. 2 Decision Boundary

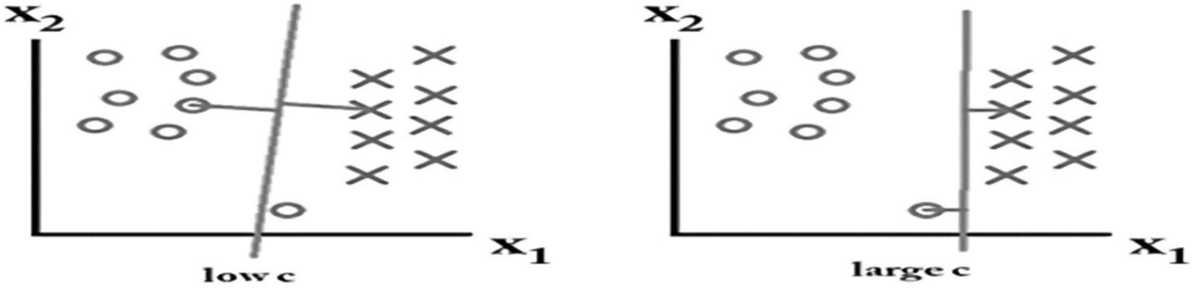


Fig. 3 Effects of *C* Parameter on SVM model fitting

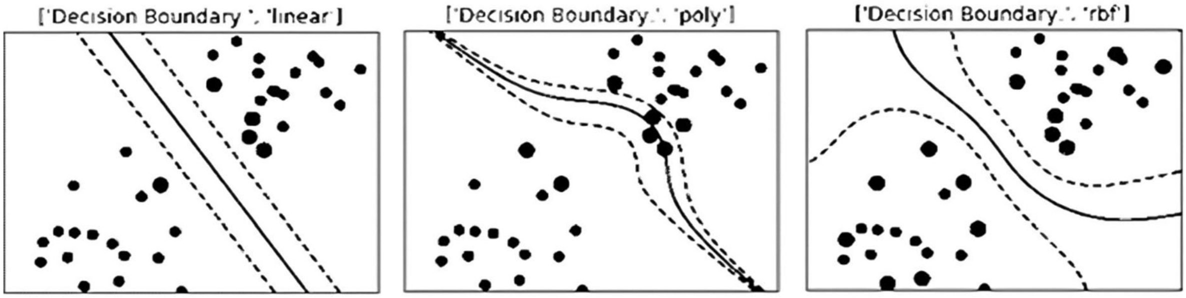


Fig. 4 Gamma Hyperparameter

low value provides more flexibility, but the model will suffer from underfitting. Figure 4 depicts the effects of Gamma. The value of Gamma exists between 0.0001 to 10. These figures are taken from our previous study [61].

3.3 Optimization of algorithm

Hyperparameter optimization is a critical component of machine learning research. Various models have been developed by researchers to optimize hyperparameters and improve the effectiveness of ML models. Some well-known techniques for enhancing the performance of SVM include Grid Search Cross-Validation, Bayesian Search Cross-Validation, Particle Swarm Optimization, GWO, among others. The GWO algorithm's computational cost is roughly proportional to the number of wolves, iterations, and problem dimensions. Although the manuscript does not directly address GWO's computational cost, its linear complexity suggests moderate computational demands compared to more complex algorithms. The study demonstrates that the use of GWO for optimizing SVR significantly enhances performance, achieving higher accuracy and lower error metrics than simpler techniques like Grid Search. Thus, the benefits of GWO justify its computational cost in this context. The specific steps for applying GWO to optimize SVR are as follows:

- Define the initial population of wolves (solutions) randomly. Each wolf's position corresponds to a set of SVR hyperparameters (C, Gamma, etc.).
- Use the SVR model with the current set of hyperparameters to evaluate the model's performance based on an objective function (e.g., minimizing the Mean Squared Error).
- Rank the wolves based on their performance, selecting the best as alpha, beta, and delta wolves.
- Update the positions of the wolves iteratively, where the new positions represent new sets of SVR hyperparameters. This step is influenced by the alpha, beta, and delta wolves, following the GWO's position update equations.
- Repeat the position updates until the stopping criterion is met (maximum iterations or a negligible change in performance).
- Apply the optimal set of SVR hyperparameters found by GWO to the testing data and evaluate the final model performance.

This detailed process would enhance clarity and guide readers through the practical implementation of GWO with SVR.

Algorithm:SVR Fine-Tuned with Grey Wolf Optimization (SVR-GWO)

Step 1: Initialize Parameters

Input: Dataset $D = \{(x_i, y_i)\}_{i=1}^{N}$ where x_i is the feature vector of sample i, and y_i is the target (heating or cooling load).
Parameters:

- C: Regularization parameter of SVR.
- Γ: Kernel coefficient for the RBF kernel.
- ϵ : Insensitivity parameter for SVR.
- α, β, δ: Leader wolves in GWO.
- **Population size n**: Number of wolves.
- **Max iterations T**: The number of iterations for optimization.

Step 2: Objective Function for SVR

- Define the objective function for SVR as:

$$\min_{w,b,\xi,\xi^*} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{N}(\xi_i + \xi^*{}_i)$$

Subject to:

$$y_i - w.\varphi(x_i) - b \le \epsilon + \xi_i$$

$$w.\varphi(x_i) + b - y_i \le \epsilon + \xi_i$$

Where ξ and ξ^* are slack variables, C is the regularization parameter, and γ is the kernel coefficient for the RBF kernel:

$$K(x_i, x_j) = \exp(-\gamma\|x_i - x_j\|^2)$$

Step 3: Initialize Grey Wolf Optimization

1. **Initialize Population**: Generate random initial positions (candidate solutions for C, γ, ϵ for n wolves.
2. **Evaluate Fitness**: Use cross-validation to evaluate each wolf's fitness based on the SVR's performance using the MAE, MSE, and RMSE metrics:

$$Fitness = \frac{1}{2}\sum_{i=1}^{N}|y_i - \hat{y}_i|$$

Where $\hat{y}_i$ is the predicted output by the SVR model.

Step 4: Update Wolf Positions (Optimization Loop)

For each iteration t=1, 2, ..., T:

1. **Sort Wolves**: Rank wolves based on fitness and assign the top three as α, β, and δ wolves.
2. **Update Positions**:
   - Update the position of each wolf based on the position of the leader wolves:

$$X(t+1) = \frac{1}{3}\left(x_\alpha(t) + x_\beta(t) + x_\delta(t)\right) - A.D$$

Where A is a random vector, and D is the distance between a wolf and the leader wolves.

3. **Boundary Check**: Ensure that the new parameters are within acceptable bounds:

$$C\epsilon[0.1, 1000]$$

$$\gamma\epsilon[0.0001, 1]$$

$$\varepsilon\epsilon[0, 1]$$

4. Evaluate New Positions: Recalculate the fitness of each wolf using the new parameters.

Step 5: Convergence Check

- Stop the algorithm if the difference between the fitness of the top wolves (α, β, δ) is below a threshold or the maximum number of iterations is reached.

Step 6: Train the Final SVR Model

- Use the best hyperparameters found by GWO (C^*, γ^*, ϵ^*) to train the final SVR model on the entire training dataset.

Step 7: Prediction

- Use the optimized SVR model to predict heating and cooling loads for new data points.

This algorithm integrates the hyperparameter tuning of SVR through GWO, leveraging the exploration and exploitation mechanisms of GWO to fine-tune SVR for energy efficiency predictions. The algorithm optimizes three key parameters—C, γ, and ε—to minimize errors such as MAE and RMSE.

3.4 Linear regression

Linear regression (LR) is a machine learning technique used to find the relationship between a dependent variable and one or more independent variables. It assumes a linear connection between these variables. If there's only one independent variable, it's called simple LR. If there are multiple independent variables, it's called multiple LR [31]. Mathematical notations of different types of LR are given using the following equations.

$$Y = \beta_0 + \beta_1 X + \varepsilon \tag{9}$$

$$Y = \beta_0 + \beta_1 X + \beta_2 X + \cdots + \beta_n X + \varepsilon \tag{10}$$

where $\boldsymbol{\beta}_0$ represents the y-intercept, $\boldsymbol{\beta}_1$ to $\boldsymbol{\beta}_n$ are the slopes of the regression line, n is the number of observations and ε is the error term accounting for the difference between the observed and predicted values. LR is widely used for predictive modeling and inference, providing interpretable coefficients that explain how much the dependent variable changes in response to a one-unit change in the independent variable.

3.5 Decision tree

A decision tree is a machine learning model that categorizes data by splitting it into smaller groups based on specific characteristics. It is used for both classification and regression tasks. In regression, decision trees serve as a non-parametric model that recursively partitions the feature space into distinct regions, aiming to minimize the variance within each partition [62]. At each internal node of the tree, the data is split based on a feature and a threshold value that optimally reduces the MSE of the target variable. The main objective of DT regression is to partition the input space into regions and assign a predicted value, typically the mean or median of the target variable in each region. Mathematically, the tree splits the feature space into M regions $R_1, R_2, \ldots, R_m$ and for each observation x in region R_m, the predicted value is computed using the following equation.

$$\hat{y}(x) = \frac{1}{|R_m|} \sum_{i \in R_m} y_i \tag{11}$$

where y_i represents the observed target values for the samples in the region is the number of observations in that region R_m and R_m is the number of observations in that region. It uses MSE to determine the best split at each node. Mathematical notation for MSE is given using the following equation.

$$MSE = \frac{1}{N} \sum_{i=1}^{N} (y_i - \overline{y})^2 \tag{12}$$

where N is the number of data points in the node, y_i is the actual target value, and $\overline{y}$ is the predicted mean value for that node. The algorithm seeks to minimize the total MSE across all nodes by selecting the split that reduces the variance the most.

3.6 Random forest

Random forests (RF) are a machine learning technique that improves the accuracy and stability of decision trees (DT) by creating a group of trees. In regression problems, each tree in the forest is trained on a different portion of the data using a method called bootstrap aggregation [56]. Furthermore, when splitting the data in each tree, only a random selection of features is considered, which helps to prevent overfitting and makes the forest more diverse. The prediction of the random forest is the average of the predictions from all individual trees, expressed as:

$$\hat{Y} = \frac{1}{T} \sum_{t=1}^{T} \hat{Y}^{(t)} \tag{13}$$

where T is the number of trees, and $\hat{\boldsymbol{Y}}^{(\boldsymbol{t})}$ is the prediction from the *t-th* tree. This aggregation reduces the variance typically associated with individual decision trees, leading to a more generalized model with a reduced risk of overfitting.

Mathematically, random forests aim to minimize the mean squared error (MSE) in regression or maximize classification accuracy by combining weak learners (individual trees) into strong learners. The randomness introduced through both data sampling and feature selection ensures that the trees are diverse, making the ensemble less prone to overfitting while maintaining high accuracy on unseen data. Random forests also provide feature importance scores, which offer insights into which variables have the most influence on the prediction, making them interpretable and useful in high-dimensional problems. However, they may require significant computational resources due to the need to train and store multiple trees, especially when the number of features or observations is large.

3.7 Deep neural network

Deep neural networks (DNNs) are machine learning models that mimic the structure of the human brain. They consist of multiple layers of interconnected nodes, like neurons [63, 64]. Neurons calculate a weighted average of their inputs and then pass the result through a non-linear activation function. Mathematically, for a single neuron in layer l, the output $a_i^{(l)}$ is given by:

$$a_i^{(l)} = \sigma\left(\sum_{j=1}^{n^{(l-1)}} w_{ij}^{(l)} a_j^{(l-1)} + b_i^l\right) \tag{14}$$

where $a_i^{(l)}$ represents the weight between neuron j in layer $l-1$ and neuron i in layer l, b_i^l is the bias term for the neuron $a_j^{(l-1)}$ is the activation from the previous layer, and σ is the activation function, such as the ReLU (Rectified Linear Unit), sigmoid, or tanh. The network predicts $\hat{Y}$ for a given input by passing the input through its layers, with each layer contributing to the final prediction. DNNs are usually trained with backpropagation. This method reduces the loss, which is often measured by MSE for regression or cross-entropy for classification. The weights and biases are adjusted to achieve this. Gradient descent is used to calculate the necessary adjustments. This involves using the chain rule to find how the loss changes with each parameter. Given a loss function $L\left(Y, \hat{Y}\right)$, the weight update is expressed as:

$$w_{ij}^l = w_{ij}^l - \eta \frac{\partial L}{\partial w_{ij}^l} \tag{15}$$

where η is the learning rate. Deep neural networks are highly effective for modeling complex, non-linear relationships, especially in tasks like image recognition, natural language processing, and time series [65] forecasting, but they require large amounts of data and computational resources to train effectively.

4 Proposed model

This part explains our suggested model and goes into detail about how we adjusted its settings and put it to use. Figure 5 shows the steps involved in our model for predicting how energy-efficient different building designs are. The objective of this study is to predict the HL, and CL of buildings, which are essential for efficient energy management and meeting the comfort requirements. The prediction problem is formulated as a regression task where the goal is to learn a function that maps a set of building features X to the corresponding heating and cooling loads.

Given a dataset $D = \{(x_i, y_i)\}_{i=1}^N$ where x_i represents the feature vector of the ith sample, and y_i is a vector containing the heating and cooling load values, the aim is to find a function f that minimizes the prediction error over D, discussed in Eq. (16)

$$f : \mathbb{R}^{\mathrm{d}} \to \mathbb{R}^2 \tag{16}$$

The function f is modeled using *SVR*, which is defined for a single output as in Eq. (17)

$$f(x) = \langle \mathrm{w}, \Phi(x) \rangle + b \tag{17}$$

where $\langle \cdot, \cdot \rangle$ denotes the dot product.

Φ: $\mathbb{R}^d \to \mathbb{R}^m$ is a feature mapping function corresponding to the chosen kernel, and w $\in \mathbb{R}^{\mathrm{m}}$ is the bias term.

The SVR aims to find w and b by solving the following optimization problem discussed in Eq. (18)

$$min_{w,b,\xi,\xi^*} \frac{1}{2}\|\mathrm{w}\|2 + \mathrm{C} \sum_{i=1}^{N} \left(\xi_i + \xi_i^*\right) \tag{18}$$

Subjects to the following constraints.

$$y_i - \langle w, \Phi(x_i) \rangle - b \le \varepsilon + \xi_i$$

$$open - angle\langle w, \Phi(x_i) \rangle + b - y_i \le \varepsilon + \xi_i^*$$

$$\xi_i, \xi_i^* \ge i = 1, \ldots, N$$

In the above equations, ξ_i and ξ_i^* are slack variables that allow for errors larger than ε. C is the regularization that controls the trade-off between the flatness of f and the amount up to which deviations larger than ε are tolerated.

To optimize the parameters of the SVR models, this study employed GWO, which iteratively updates a population of solutions (wolves) to converge toward the optimal solution. The position of each wolf X_i in the search space is updated based on the positions of the alpha (α), beta (β), and delta (δ) wolves, which correspond to the best solutions found so far. The position update equations for the i^{th} wolf is given by the following Eq. (19).

$$X_i^{(t+1)} = \frac{1}{3}(X_\alpha^{(t)} + X_\beta^{(t)} + X_\delta^{(t)}) - A \bullet D \tag{19}$$

where t indicates the current iteration, A is a coefficient vector and D is the distance vector between the ith wolf and the alpha, beta, and delta wolf.

The GWO algorithm iteratively updates the positions of the wolves until convergence criteria are met, such as reaching a maximum number of iterations or achieving a sufficiently small change in the positions of the alpha, beta, or delta wolves.

Table 4 presents the initial hyperparameters for SVM model optimization. The hyperparameters include

C: Regularization parameter with values of 1, 5, 10, 100, and 1000;

Gamma: Kernel coefficient for 'rbf' (radial basis function) kernel, with values of 0.1, 0.01, 0.001, and 0.0001.

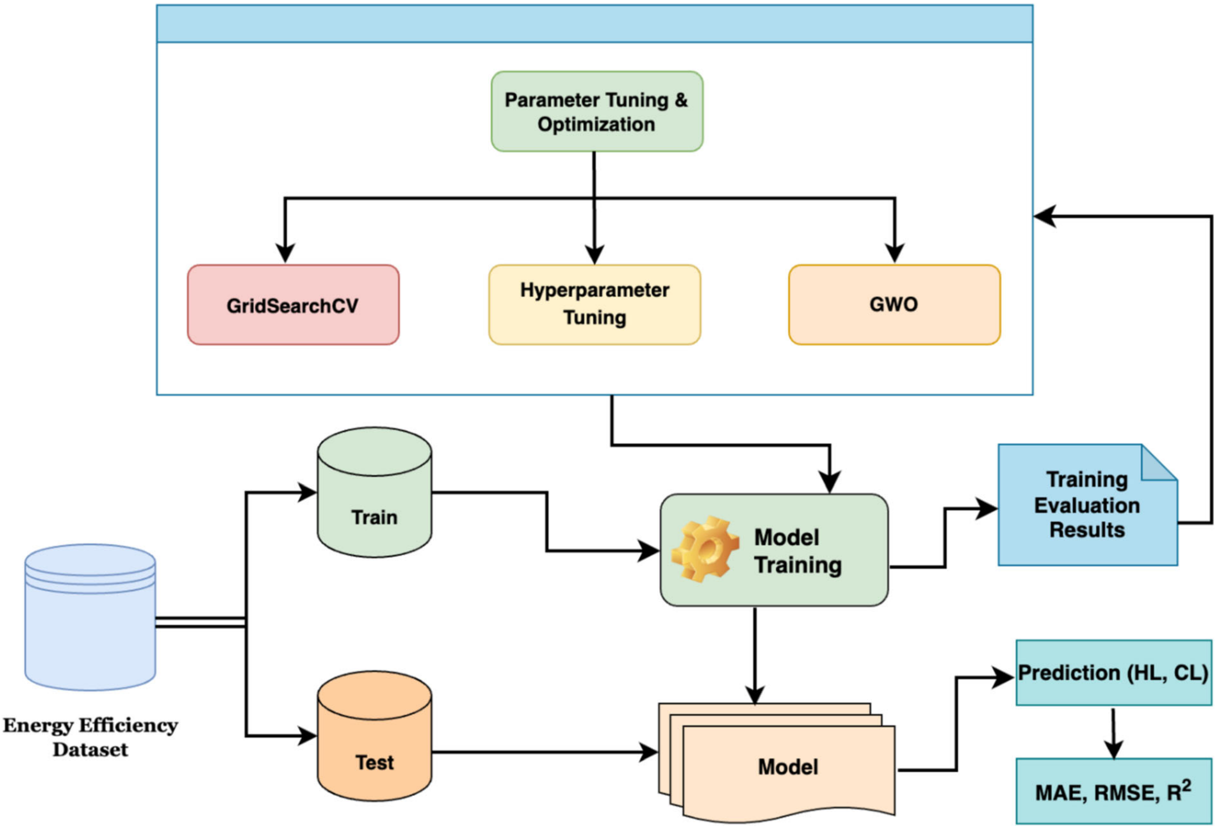


Fig. 5 Workflow of the proposed model

Table 4 Initial hyperparameters

Hyperparameter	Values
C	1, 5, 10, 100, 1000
Gamma	0.1, 0.01, 0.001, 0.0001
Kernel	RBF

The choice of hyperparameters (C, Gamma, and RBF kernel) was primarily driven by the need to handle the non-linear patterns observed in the dataset. After conducting preliminary experiments with multiple kernels (linear, polynomial, and RBF), the RBF kernel consistently provided better results due to its ability to handle non-linearity more effectively than the linear or polynomial alternatives. This observation aligns with similar studies in literature, where RBF is often the preferred choice for complex, non-linear problems.

The system (see Fig. 5) takes an energy efficiency dataset, tunes hyperparameters using Grid Search Cross-Validation and GWO, trains a model, and evaluates it using MSE, RMSE, and R-squared on test data, aiming to optimize model performance.

5 Experimental settings

In this section, experimental settings like data set description, and evaluation metrics that have been utilized to measure the performance of the proposed model are discussed.

5.1 Data description

The dataset used in the study is the "*Energy Efficiency Data Set*" from the UCI ML Repository (https://archive.ics.uci.edu/dataset/242/energy+efficiency) [8]. It is designed for energy efficiency research, particularly for predicting the heating and cooling loads of buildings based on various architectural features. The data is developed using the Ecotect by providing 12 different simulations of building architecture. The architecture of each building prototype has different glazing distribution and orientation. These various setting of glazing area is provided in functions of the afore-mentioned characteristics and generate 768 observations with 8 features, and 2 values to make an estimation. Table 4 presents the clear meaning of 8 input variables denoted by *X* and 2 output variables depicted by *y1* and *y2*. Furthermore, all the buildings have the same volume (771.75 m^3) and material used to construct them.

Data also simulated all the shapes by considering that buildings are constructed in the Athens, Greece area with a hypothesis of seven people inside with sedentary activity accounted to approximately 70W. The other internal parameters of buildings are kept equal e.g. humidity, airspeed, and strength of lighting are 60%, 0.30 m/s, and 300 Lux, respectively. After fetching the data, we divided it into two parts: training (80%) and testing (20%). Training data is used to tune the parameters and train the models while testing data is used to evaluate the performance of the proposed model.

Table 5 describes the input and output variables for a system, likely related to building energy efficiency. The input variables (X1–X8) represent various building characteristics like size, shape, and materials (compactness, floor area, wall area, etc.), while the output variables (y1 and y2) represent the building's energy consumption for heating and cooling.

When it comes to a building's energy efficiency, relative compactness (R_c) is one of the most significant factors to consider. In Table 4 we have mentioned 8 parameters among which X1 denotes the relative compactness. Additionally, the shape of the building can affect the increase or decrease of the energy required to heat or cool the occupied space. Relative compactness can be calculated by the total of all surfaces of its envelope, divided by the total volume of its heating [66]. Figure 6 presents 12 different shapes of buildings and their relative compactness.

If we denote L_b as the building shape factor, V_b as the volume of the building, and S_i as the sum of all heat loss surfaces that are in contact with the exterior, ground, or adjacent non-heated spaces, then R_C can be calculated as given in Eqs. (20) and (21).

Table 5 Description of the input and output variables

Description	Variable
Relative compactness	X1
Floor area	X2
Wall area	X3
Roof area	X4
Overall height	X5
Orientation	X6
Glazing area	X7
Glazing area distribution	X8
Heating load	y1
Cooling load	y2

$$L_b = V_b / \sum_{i=1}^{n} S_i \quad \text{and} \quad L_b = R_c \cdot V_b^{0.66} \cdot 6^{-1} \tag{20}$$

$$R_c = 6 \times V_b^{0.66} \times \sum_{i=1}^{n} S_i^{-1} \tag{21}$$

The above equation shows that the degree of compactness does not depend on whether or how much the building is insulted. Orientation also doesn't work here. The geometry of the thermal envelope is all that matters.

Other features of the building such as Floor area, Wall area, Roof area, and Overall height are presented in Fig. 7.

5.2 Benchmark models

It has been discussed that researchers have used various machine learning for similar kinds of jobs. This study implements and compares the performance of different ML models like LR, DT, RF, DNN, and SVR. It has been observed that SVR performed better than all the conventional approaches including DNN. Support Vector Machine (SVM) can be used for both classification and regression types of problems. Since this study deals with HL and CL which is nothing but multivariate regression analysis, it uses SVR and applies GWO for optimization of hyperparameters.

5.3 Performance measurement

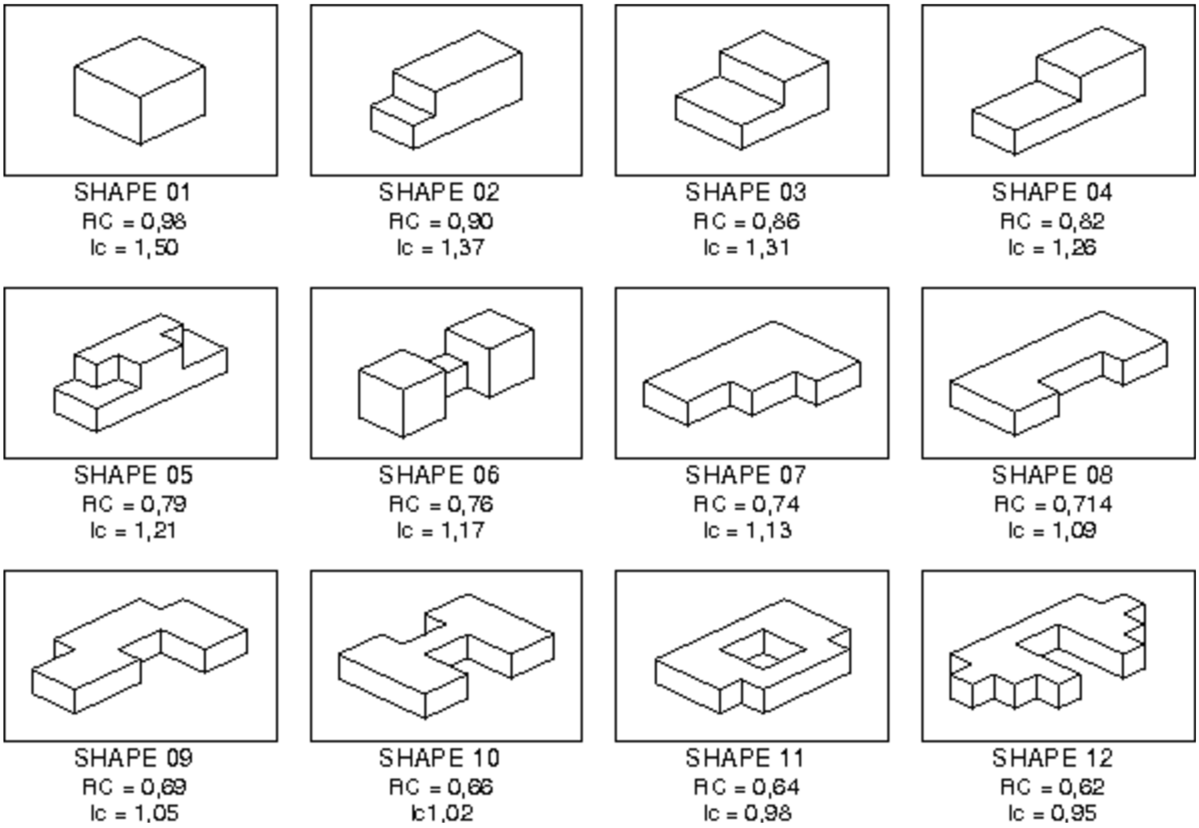
Evaluation and comparison are important aspects of establishing the supremacy of any research [68, 69]. To demonstrate the superiority of the proposed model, its performance is evaluated against established benchmarks using metrics such as mean absolute error, mean square error, root mean square error, and R^2. Furthermore, to get an insight into the properties of the data, mean and standard deviation (SD) are computed. The mathematical notations for the evaluation metrics used in this study are presented in the following equations, from Eqs. (22) to (27).

$$MAE = \frac{1}{n}\sum_{i=1}^{n}(\mathrm{Load}_{\mathrm{Predicted}} - \mathrm{Load}_{\mathrm{Actual}}) \tag{22}$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(\mathrm{Load}_{\mathrm{Predicted}} - \mathrm{Load}_{\mathrm{Actual}})^2 \tag{23}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\mathrm{Load}_{\mathrm{Predicted}} - \mathrm{Load}_{\mathrm{Actual}})^2} \tag{24}$$

$$R^2 = 1 - \frac{Unexplained\ Variation}{Total\ Variation} \tag{25}$$


SHAPE 01
RC = 0,98
Ic = 1,50
SHAPE 02
RC = 0,90
Ic = 1,37
SHAPE 03
RC = 0,86
Ic = 1,31
SHAPE 04
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SHAPE 05
RC = 0,79
Ic = 1,21
SHAPE 06
RC = 0,76
Ic = 1,17
SHAPE 07
RC = 0,74
Ic = 1,13
SHAPE 08
RC = 0,714
Ic = 1,09
SHAPE 09
RC = 0,69
Ic = 1,05
SHAPE 10
RC = 0,66
Ic1,02
SHAPE 11
RC = 0,64
Ic = 0,98
SHAPE 12
RC = 0,62
Ic = 0,95

Fig. 6 Various building structures with varying levels of relative compactness (R_c) [67]

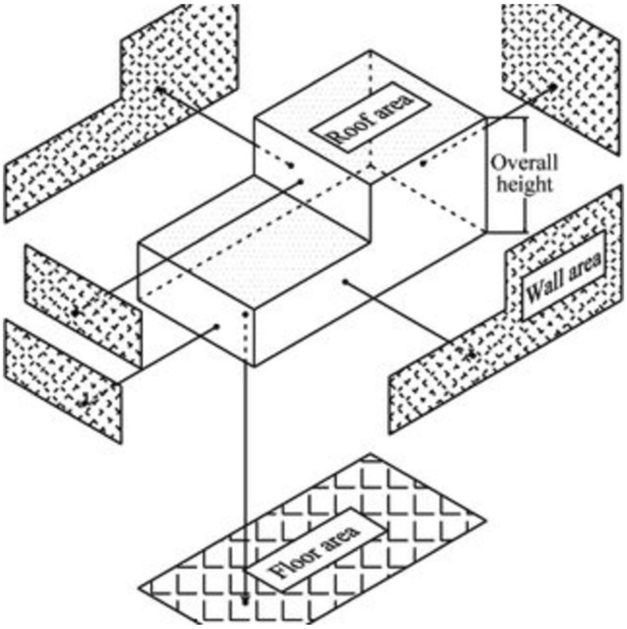

Roof area
Overall height
Wall area
Floor area

Fig. 7 Orientation of building and different features [8]

$$Mean = \frac{\sum_i^N X_i}{N} \tag{26}$$

$$theSD = \sqrt[2]{\frac{\sum_i^N \left(X_i - \overline{X}\right)^2}{N}} \tag{27}$$

6 Result and discussion

This section presents and discusses the results of the proposed model and other benchmark models obtained on different evaluation metrics. The proposed model was implemented on the dataset obtained from the UCI repository. Initially, the data was divided into training (614 cases) and testing (154 cases) groups. Subsequently, the model was constructed, and its effectiveness was assessed by comparing it to four other models. The literature review suggests that the performance of SVR was the best among all implemented models. This study optimized the parameter of SVR using GWO which further improved the performance of the SVR model.

6.1 Numerical computation

The findings in Table 6 show that the proposed SVR-GWO model significantly outperforms all other models in predicting cooling loads. The R^2 value for SVR-GWO is the

Table 6 Performance comparison for cooling load

Source	Model	RMSE	MAE	R^2
Lu et al. [69]	DNN	3.10	2.314	0.896
	AutoML	0.97	1.324	0.989
Roy et al. [30]	Gaussian Process	3.28	2.388	0.882
	LR	2.95	2.146	0.903
	ANN	3.11	2.307	0.892
	RBF Network	4.36	3.133	0.787
	SVM	3.23	2.370	0.884
Le et al. [70]	PSO-XGBoost	NC	NC	NC
	XGBoost	NC	NC	NC
	GP	NC	NC	NC
	CART	NC	NC	NC
Mehdizadeh et al. [71]	RF	1.95	1.318	0.958
	LR	3.19	2.26	0.888
Zheng et al. [72]	LM-MLP	3.234	2.293	0.888
	MFO-MLP	2.747	1.941	0.917
	OIO-MLP	2.963	2.054	0.903
	SCE-MLP	2.546	1.845	0.928
Proposed	**SVR-Grid**	**1.52**	**0.993**	**0.974**
	SVR-GWO	**1.39**	**0.838**	**0.979**

The significance of bold highlights the results achieved by our proposed model, indicating its superior performance compared to other models or benchmarks presented

Table 7 Performance comparison for heating load

References	Model	RMSE	MAE	R^2
Lu et al. [69]	DNN	1.95	1.383	0.96
	RF	1.03	0.557	0.978
	AutoML	0.60	1.34	0.996
Roy et al. [30]	Gaussian Process	3.173	2.500	0.901
	ANN	1.105	1.085	0.988
	RBF Network	0.458	0.353	0.804
	SVM	2.958	2.279	0.913
Le et al. [70]	PSO-XGBoost	1.124	0.615	0.990
	XGBoost	1.651	0.720	0.977
	GP	1.632	0.798	0.978
	CART	1.779	0.773	0.973
Mehdizadeh et al. [71]	LR	2.929	2.01	0.915
Zheng et al. [72]	LM-MLP	NC	NC	NC
	MFO-MLP	NC	NC	NC
	OIO-MLP	NC	NC	NC
	SCE-MLP	NC	NC	NC
Proposed	**SVR-Grid**	**0.5952**	**0.4319**	**0.996**
	SVR-GWO	**0.4852**	**0.3422**	**0.997**

The significance of bold highlights the results achieved by our proposed model, indicating its superior performance compared to other models or benchmarks presented

highest at 0.979, surpassing models like DNN (R^2: 0.896), Gaussian Process (R^2: 0.882), SVM (R^2: 0.884), and ANN (R^2: 0.892). Even models that performed relatively well, such as RF (R^2: 0.958) and the SCE-MLP model (R^2: 0.928), do not reach the accuracy levels of SVR-GWO. Additionally, SVR-GWO achieves the lowest RMSE (1.39) and MAE (0.838), indicating a superior ability to minimize prediction errors compared to other models, including Linear Regression (R^2: 0.903, RMSE: 2.95), OIO-MLP (R^2: 0.903, RMSE: 2.963), and LM-MLP (R^2: 0.888, RMSE: 3.234). The model's enhanced performance is due to the use of GWO, which effectively optimizes the SVR parameters to capture non-linear relationships more efficiently than traditional methods. The combination of lower error metrics and higher predictive accuracy makes SVR-GWO a robust solution for cooling load prediction, highlighting its suitability for practical applications in energy-efficient building design, where precise predictions can lead to improved HVAC system sizing and significant cost savings.

Based on Table 7, the proposed SVR-GWO model demonstrates remarkable performance in predicting the heating load. It achieves the highest R^2 value of 0.997, surpassing all other models, including AutoML (R^2: 0.996), PSO-XGBoost (R^2: 0.990), and RF (R^2: 0.978). The SVR-GWO model also delivers the lowest RMSE at 0.4852 and MAE at 0.3422, indicating superior accuracy in minimizing prediction errors. This performance is significantly better than models such as SVM (R^2: 0.913) and LR (R^2: 0.915), which show much higher error rates.

The AutoML model by Lu et al. comes close in performance but still falls short in both RMSE (0.60) and MAE (1.34), indicating that while automated hyperparameter tuning provides robust results, the SVR-GWO model benefits from the precise optimization capability of GWO. Additionally, models like XGBoost and CART perform reasonably well but still lag, with R^2 values of 0.977 and 0.973, respectively.

What makes the SVR-GWO model particularly effective is its ability to optimize SVR parameters to capture the complex relationships in the heating load data, resulting in both lower prediction error and higher accuracy. This makes SVR-GWO especially valuable for energy-efficient building design, where accurate heating load predictions are crucial for optimal HVAC system design, leading to better energy management and cost savings in real-world applications.

In Figs. 8 and 9, a performance comparison is presented for different machine learning models predicting the heating and cooling loads, respectively. The models are evaluated based on three key metrics: Root Mean Squared

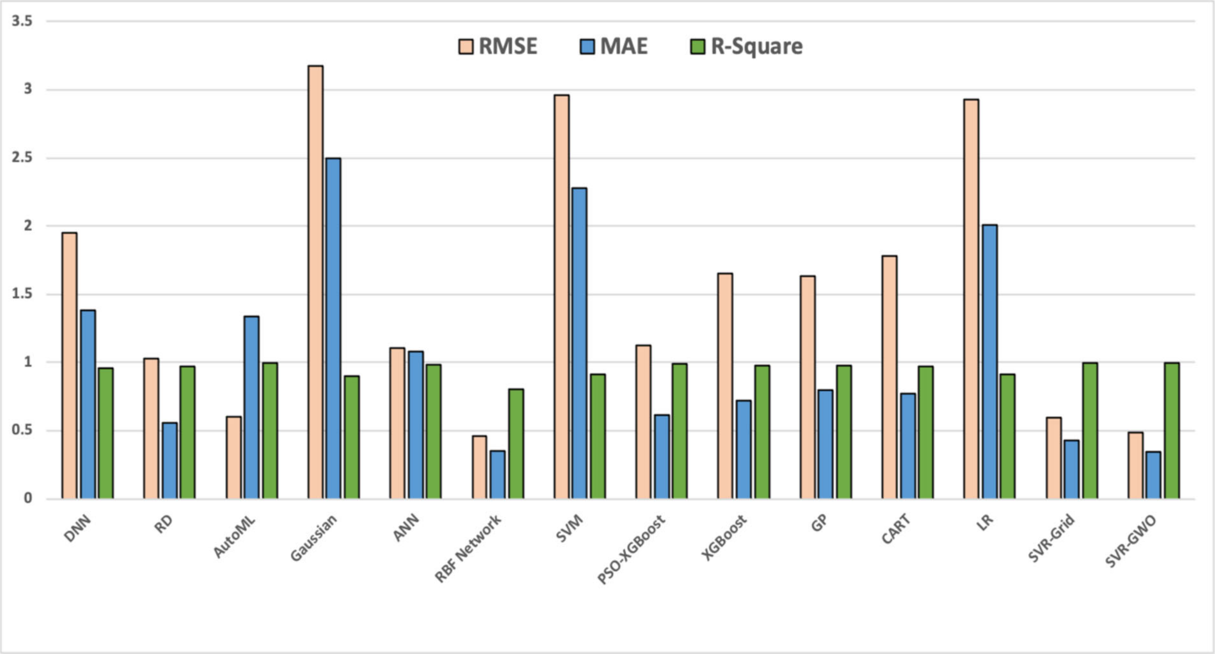


Fig. 8 Performance comparison for heating load

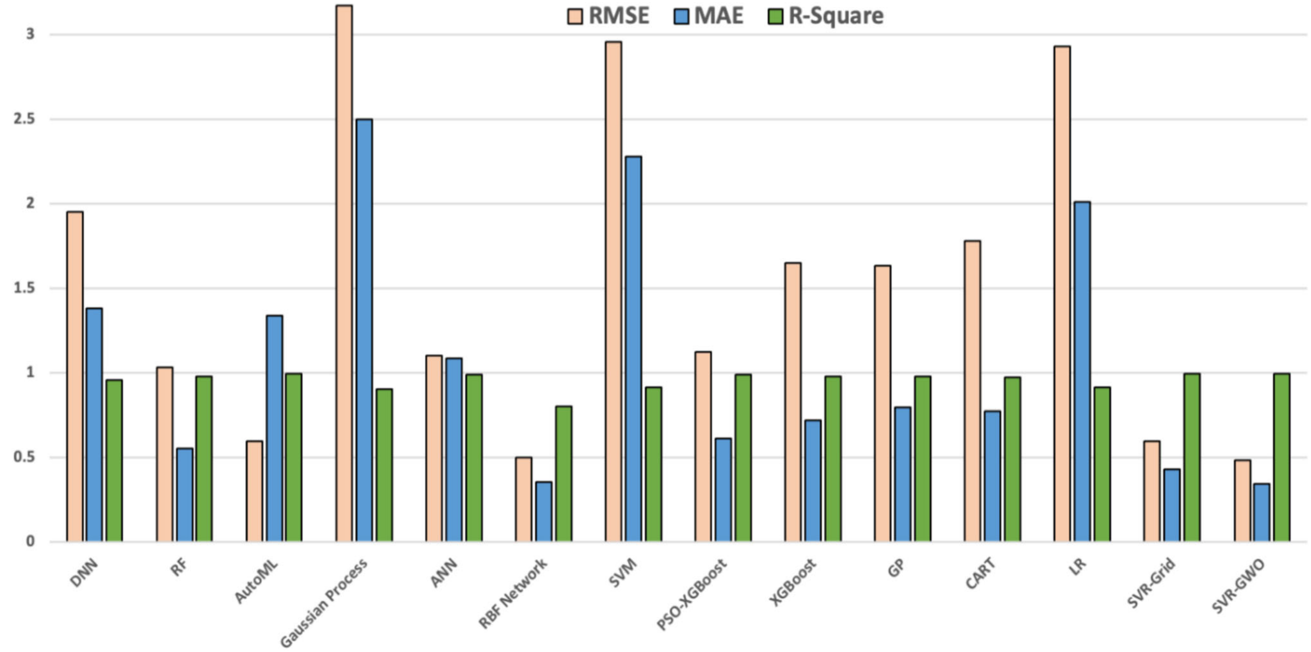


Fig. 9 Performance comparison for cooling load

Error (RMSE), Mean Absolute Error (MAE), and R-Squared (R^2).

6.2 Visual interpretation for heating and cooling

To evaluate the performance of the model, we constructed several graphical representations, which are explained hereafter.

6.2.1 Box plot

Figure 10 box plots show that the SVR-GWO model accurately forecasts heating and cooling loads. The median of both actual and predicted loads is extremely similar, revealing that the model's predictions are accurate. The model's predictions have a larger range than the actual

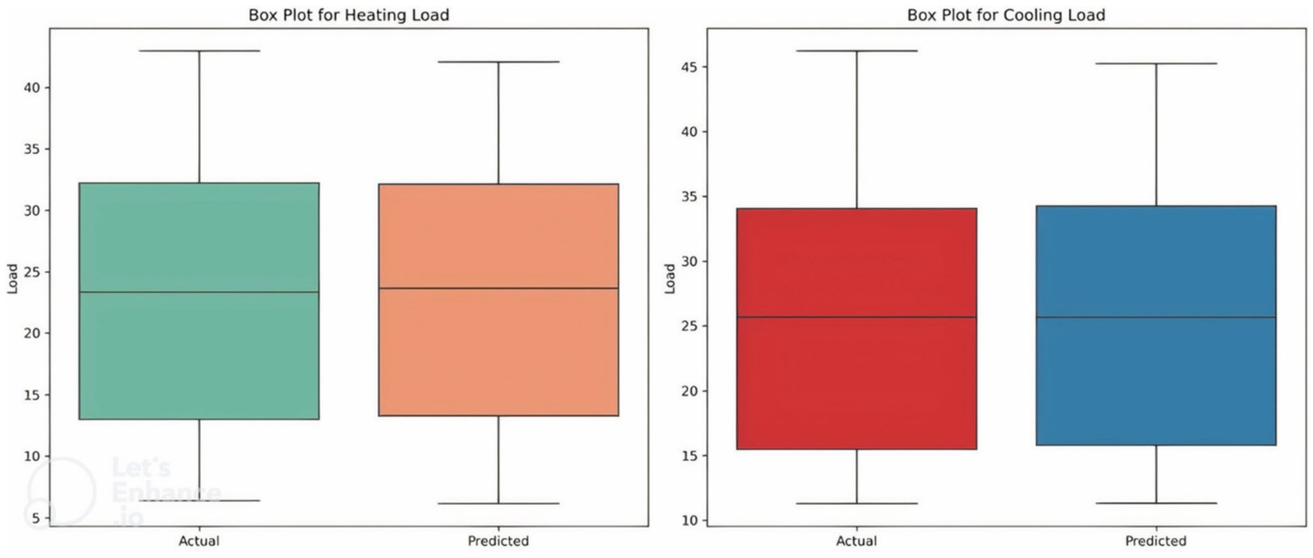


Fig. 10 Box plot of actual and predicted heating and cooling loads

measured data, showing its ability to forecast numerous instances effectively.

6.2.2 Residual plots

The residual plots (Fig. 11) show that the model's predictions for the heating load are very close to the actual values, with most errors being small and centered around zero, indicating high accuracy. For the cooling load, while the predictions are a bit more scattered, the model still gives acceptable results, especially when we consider how challenging it is to make these predictions.

6.2.3 Density plot

For both heating and cooling, the density plots show a strong agreement between the model's forecasts and the real-world data. The observed correlation indicates that the model effectively predicts the dispersed nature of the data and the distribution of most values.

Figures 12 and 13 show separate density plots comparing the actual and predicted cooling and heating loads, respectively. In both figures, the x-axis represents the load (in kWh), and the y-axis represents the density (or probability). Figures 11 and 12 compare the actual and predicted cooling and heating loads, respectively. The green and blue lines in Fig. 11 represent the actual and predicted cooling

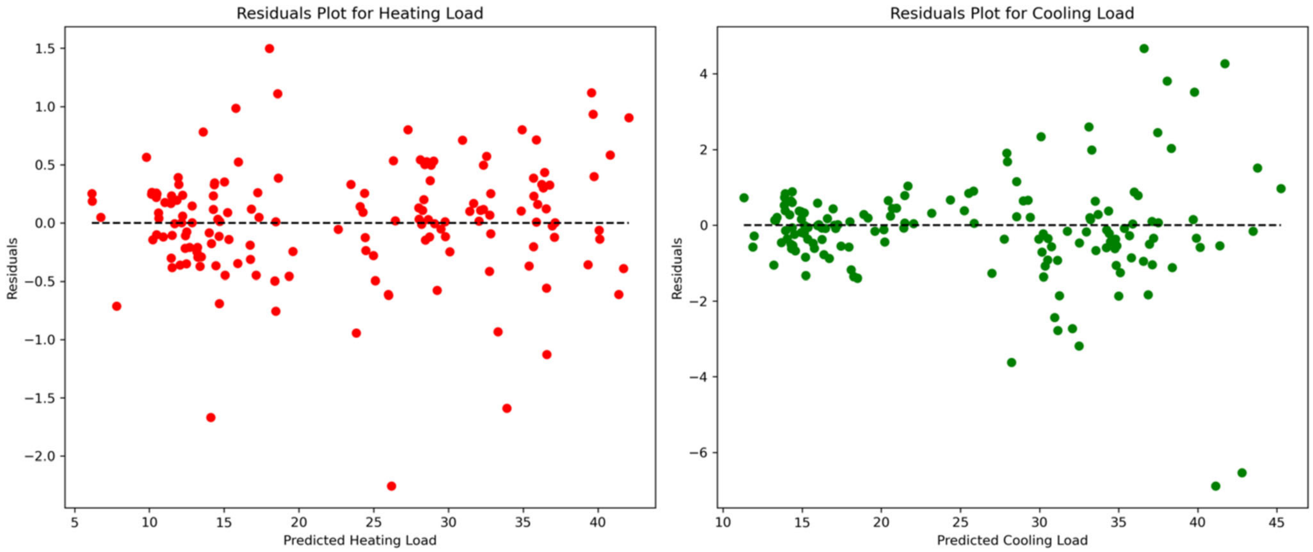


Fig. 11 Residuals plot for heating and cooling loads

Fig. 12 Density plot comparison of actual and predicted cooling load

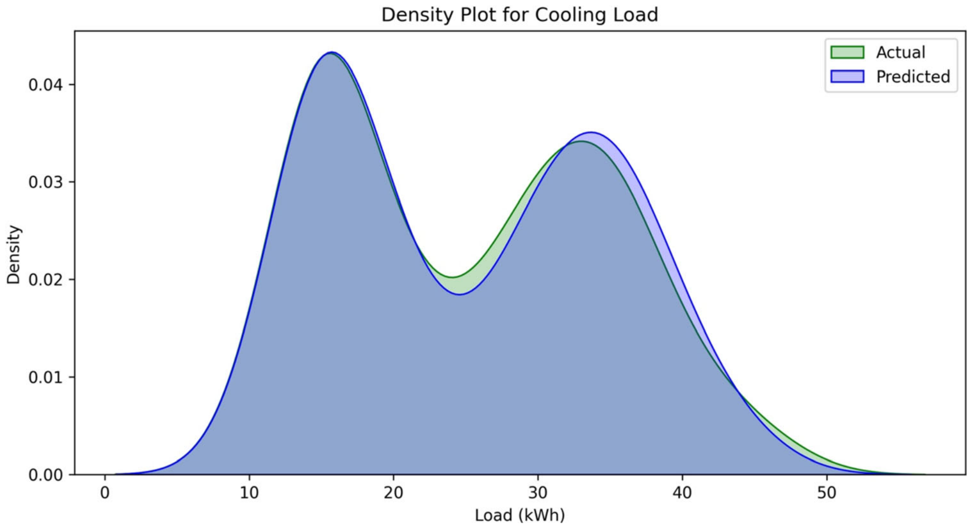
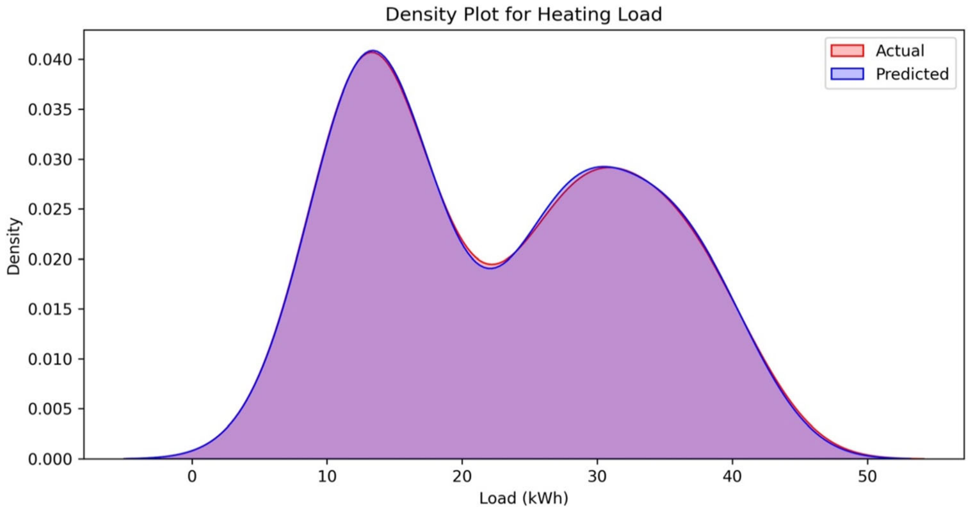


Fig. 13 Density plot comparison of actual and predicted heating loads



loads, while the red and blue lines in Fig. 12 show the actual and predicted heating loads. The close alignment of these curves indicates the model's effectiveness in predicting both heating and cooling requirements (Fig. 13).

6.2.4 Actual vs. predicted load line plots

Figure 14 presents two scatter plots that contrast the actual and predicted heating and cooling loads. The x-axis displays the actual load, while the y-axis indicates the predicted load. The red dots in the left plot represent the heating load data, and the green dots in the right plot represent the cooling load data. The dashed line represents a perfect prediction, where the actual and predicted loads would be equal. The closeness of the dots to this line indicates the accuracy of the prediction model. In general, the model appears to be more accurate for predicting heating load than cooling load.

The SVR-GWO model demonstrated superior performance in forecasting both heating and cooling loads, exhibiting minimal prediction errors.

6.3 Discussions

The comparative analysis of various machine learning models for heating and cooling load prediction demonstrates that the SVR-GWO model significantly outperforms traditional and deep learning approaches. The SVR-GWO model consistently outperforms other models in predicting heating and cooling loads, as demonstrated by its superior results in RMSE, MAE, and R^2 metrics. This confirms its reliability and adaptability in various scenarios.

The SVR-GWO model significantly outperformed other methods, including the Gaussian Process, SVM, and even SVR-Grid, in predicting heating load. It achieved an exceptional RMSE of 0.4852, MAE of 0.3422, and an R^2 of 0.9977. The improvement can be attributed to GWO's

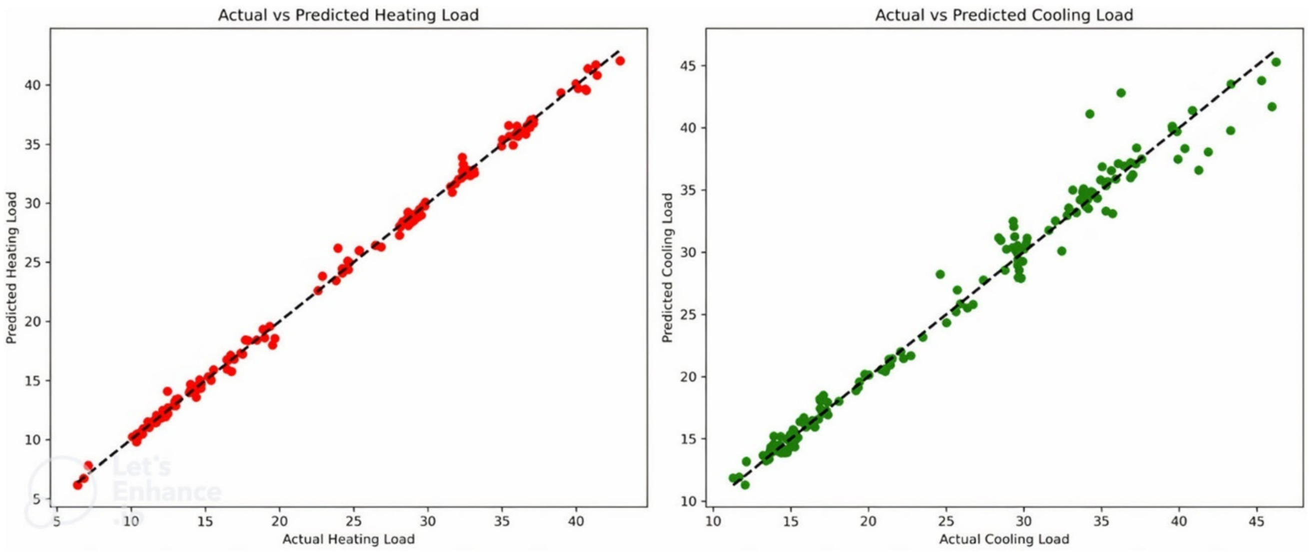


Fig. 14 Line plot of actual versus predicted loads

efficient hyperparameter optimization process, which tunes SVR's kernel parameters to better capture complex relationships in the data. The ability to handle non-linear patterns inherent in building heating load data gives SVR-GWO a clear advantage over models like DNN, RF, and RBF Networks, which yield higher error rates and lower R^2 values. Similarly, for cooling load prediction, SVR-GWO again demonstrated its superiority, with an RMSE of 1.3937, MAE of 0.8387, and R^2 of 0.9790. The optimized SVR model consistently produced more accurate predictions compared to the SVR-Grid and other benchmark models, such as GPR, GBM, and RF. The results indicate that the optimization offered by GWO plays a critical role in fine-tuning the SVR model for highly accurate cooling load prediction, a task known for its non-linear and dynamic characteristics.

The significant improvement observed in SVR-GWO's performance is a direct result of the GWO ability to explore the hyperparameter space more effectively than traditional grid search methods. GWO prevents the SVR model from getting trapped in local minima, allowing it to converge on an optimal solution. This, in turn, translates into lower error metrics and more accurate predictions for both heating and cooling loads. Traditional models like the Gaussian Process, SVM, and ANN delivered satisfactory results but were generally outperformed by SVR-GWO, particularly in minimizing prediction errors. While DNN and RF showed some promise, their inability to capture the complex, non-linear relationships in the data as effectively as SVR-GWO indicates that advanced optimization techniques like GWO are crucial for improving model accuracy in this domain. These findings suggest that moving forward, employing metaheuristic optimization methods should become a standard practice when developing machine learning models for energy load prediction.

Appendix A contains the actual and predicted dataset, sourced from the UCI Machine Learning Repository, which includes key building parameters and results from model predictions. The dataset involves various input variables such as Relative Compactness (X1), Floor Area (X2), Wall Area (X3), Roof Area (X4), Overall Height (X5), Orientation (X6), Glazing Area (X7), and Glazing Area Distribution (X8). The output variables are Heating Load (y1) and Cooling Load (y2). Model performance was evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2), offering a comprehensive analysis of prediction accuracy and model efficiency.

7 Conclusion

Electricity is a vital energy source that is challenging to store, underscoring the importance of optimizing electricity consumption in buildings. Research indicates that the primary contributors to electricity usage in residential buildings are heating and cooling systems. Consequently, there is an increasing emphasis on designing energy-efficient buildings that minimize the need for heating and cooling. To address these concerns, this paper proposed SVR-GWO for enhancing energy efficiency and building prototype architecture. The proposed model was employed on a dataset consisting of various building prototypes to predict the cooling and heating demands before the construction phase begins. The proposed model, when tested on the UCI dataset, achieved impressive results for this task. Our results reveal an impressive accuracy level, with an r-score of 99.77% for heating load predictions and 97.90% for cooling load estimations. Additionally, the model exhibits lower error metrics compared to other methods, indicating its reliability in estimating cooling and heating loads. This study is important because it provides an initial understanding of a building's energy requirements, allowing architects and builders to make informed adjustments in the design to improve energy efficiency. It is crucial to acknowledge that this study was carried out in a controlled environment, assuming homogeneity in terms of geographical location, weather conditions, and construction materials among all prototypes. Here are some possible future scopes that can be addressed:

- Adding more features to the dataset could enhance the estimation method significantly.
- A broader and more reliable method would be applicable in diverse environmental and architectural contexts.
- Moreover, future research could explore how adapting the model to different environmental and architectural factors may influence its accuracy, ensuring more robust applications beyond the Athens-based simulations used in this study.

Appendix A: Actual and Predicted Datasets

Actual heating load	Predicted heating load	Actual cooling load	Predicted cooling load
16.47	16.28619039	16.9	17.75647556
13.17	13.5217462	16.39	16.58607045
32.82	32.87057262	32.78	32.75364
41.32	41.84365055	46.23	45.27278397
16.69	17.10118098	19.76	20.27642804
23.8	23.33938037	24.61	28.80313739
24.31	24.45379317	25.63	25.56744018
28.67	28.5577854	29.62	29.493621
19.52	17.97747944	22.72	21.48899657
28.07	28.23337872	34.14	33.49994759
18.46	18.52804671	21.53	21.50269444
33.08	32.76276034	34.11	34.36212416
29.79	29.76613866	29.92	29.95829823
10.37	10.43073144	13.44	13.3538268
17.69	18.13548493	20.82	20.43445393
36.95	37.13652094	36.87	37.08740502
36.64	36.36029541	37.01	36.20665215
12.3	11.94841249	15.24	14.38566563
14.7	14.37226551	16.77	16.51540506
32.38	33.1072694	33.62	34.44316779
28.67	28.35888858	32.43	31.04869531
41.4	40.74798733	45.29	44.43401204
11.7	11.51037752	13.88	14.61694506
36.86	36.17397028	34.25	41.00160141
15.37	14.99636252	19.18	18.83324263
35.89	36.08440617	43.3	39.76115679
36.57	36.35007852	35.39	36.23880695
28.91	28.51655225	29.64	28.15351274
17.5	17.2352218	21.13	20.71207467
13.99	14.6936826	14.35	15.10403603
10.42	10.24012956	13.39	13.46633921
12.3	12.43987192	15.44	15.10581576
16.93	16.75231377	20.03	20.23429812
26.84	26.34722411	30.17	29.90770027
35.94	35.45830208	43.33	42.80880889
28.18	28.19598947	30.18	30.20053105
32.72	32.74932699	33.78	34.67861621
36.9	36.73882758	34.43	34.82480754
36.06	35.82810014	34.94	36.47571454
10.7	10.53896433	13.87	14.0228912
28.15	27.9593248	30	30.44786486
12.32	11.97535639	14.92	14.34505844
6.37	6.137418949	11.29	11.52062456
31.84	31.67713036	31.6	31.22357879
6.79	6.700922139	12.05	11.33944917
6.4	6.136718786	11.67	11.43064739
7.1	7.777107951	12.14	13.43504329
10.8	10.72060708	14.28	14.01138462
28.66	29.0809019	34.73	35.39306762
32.46	32.47239949	35.62	36.18249663
34.95	35.22045737	35.04	37.26545668
12.93	13.33146142	15.83	16.8218438
13	13.20962699	15.87	16.08933772
40.12	39.71544024	37.26	38.81610334
12.27	12.13469188	14.97	14.25648447
15.32	15.23130539	19.42	19.55763975
12.45	12.08775421	15.1	14.84360544
24.6	25.21626816	29.31	32.26367189
40.68	39.86387541	40.36	38.43767899
35.99	36.70649095	36.07	36.97466879
14.37	13.74949982	16.54	15.97559979
16.54	16.27448052	16.88	17.53111044
13.02	13.19392765	16.06	15.86823798
22.89	23.93242384	28.88	29.18464323
28.61	28.66020835	30.2	30.93698915
37.1	36.63422555	35.29	34.29442435
35.45	36.80057306	41.86	38.08615735
11.68	11.72586081	13.9	15.13483247
13	12.87415387	14.47	14.25073064
12.03	11.7945522	13.79	14.26997565

Actual heating load	Predicted heating load	Actual cooling load	Predicted cooling load
29.14	28.83619141	29.58	29.87868614
32.84	32.3249674	32.88	33.56332841
12.47	12.73168907	15.14	15.88280286
24.23	24.25820112	25.02	24.28987356
31.64	30.92151647	36.86	34.66091757
12.16	12.07059206	15.18	14.74012765
29.06	28.98534056	33.84	34.73713983
24.7	25.1662462	28.77	29.1389681
32.26	32.27297389	33.37	33.36626711
19.34	19.20465521	23.49	22.23161053
24.38	24.3118361	25.91	25.88759823
40	39.61133392	36.26	43.38777685
10.68	10.5736248	14.3	14.01880548
13.91	14.11512477	14.89	15.11550432
11.6	11.32294241	13.7	14.27732773
12.43	14.03320409	15.59	16.44716163
26.46	26.41792363	27.4	27.60814666
36.13	36.15952396	37.58	37.47367654
37.12	37.04499275	35.28	35.66413434
17.37	17.21997634	21.08	20.34454539
15.18	15.37539329	19.34	19.2747709
25.38	25.90761017	26.72	25.42596326
11.13	11.2933534	14.61	13.63699882
12.95	13.30830389	15.95	16.12671845
36.59	35.90268327	36.44	36.92290393
12.86	12.82262697	16.17	16.25451621
25.36	26.15633189	32.04	31.56450128
28.31	28.54620742	34.15	34.48388864
10.66	10.63333403	13.67	14.00416638
10.71	10.48697358	13.8	13.69034109
40.78	40.99762932	39.55	40.07978288
28.37	28.28320667	29.28	30.03381219
10.36	10.08873302	13.43	13.33487996
29.83	29.62318806	29.82	29.34860697
29.43	29.39403003	28.38	31.55831962
16.76	15.93408613	17.36	17.81444694
22.58	22.76010255	28.51	32.63282474
16.44	16.7924671	17.1	17.89418346
14.08	14.36415297	17.02	16.99882534
12.41	12.3830025	15.28	14.98180706
33.12	32.68675788	34.17	34.25080542
32.31	32.42991856	29.69	28.28702979
24.63	24.43912764	26.37	25.86930268
14.72	14.70658256	18.1	17.95873153
11.69	12.17875378	14.76	15.00138023
14.66	14.37244019	17.37	16.8936385
11.2	11.17029246	14.73	14.14830561
10.39	10.14058949	13.6	13.47046415
12.12	12.54423284	14.97	15.24859049
19.68	18.95491229	29.6	29.24154283
29.06	28.55576551	29.34	32.85763178
11.16	11.54200283	14.39	13.93530359
23.93	25.79088643	29.4	30.97943807
14.6	15.12330781	15.3	15.32202501
29.68	29.65614861	29.44	30.37305798
17.88	18.32345028	21.4	20.81106177
14.53	14.57468434	16.9	17.20226656
35.48	35.46418321	41.26	37.06474264
24.23	24.24590749	25.72	26.72295452
39.97	40.35415864	40.85	41.17450537
15.55	15.75113012	21.33	21.58707707
32.31	33.8886278	34.05	34.64328682
14.6	14.66196643	15.14	15.24622246
13.97	14.19931172	16.08	16.1800813
14.4	14.23737555	17.27	17.09899635
19	18.65223988	22.25	21.54101322
32.06	32.1475139	35.71	33.29112141
42.96	42.01976699	39.56	40.48597091
14.5	14.16886589	17.03	16.9963832
18.88	19.25280487	22.07	22.00316099
10.75	10.46976637	14.27	14.14110341
40.6	39.77406174	39.85	39.52722598
28.55	28.51533674	29.59	30.28633529
31.53	31.51971014	37.19	36.93795136
35.73	35.20283443	39.92	37.70369141
11.42	11.49351271	14.75	14.02583521
10.07	10.19402617	13.21	13.67678601
28.09	27.24248544	34.33	34.38479155
35.01	35.46456281	33.14	35.48565922
38.98	39.31853341	45.97	40.57364681
36.66	36.64224347	35.92	36.3494153
29.34	28.87252591	33.37	32.94020481
29.54	28.98590271	33.98	33.32659197
12.17	12.31027809	15.2	15.60937045

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Declarations

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